

Extended essay

The Effect of Honey Type and Temperature on Its Antibacterial Properties

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Research Question: How do honey type (acacia, buckwheat, and honeydew) and temperature (5 °C, 20 °C, and 60 °C) affect hydrogen peroxide concentration measured by potassium permanganate titration, and how might these changes influence antibacterial effectiveness against *Staphylococcus aureus* based on literature values?

Subject: Biology

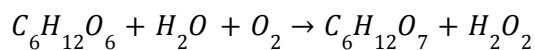
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Introduction

As an individual who lives in a household where the parents whenever someone gets ill prepare a hot tea with honey in order to help cure due to it being a natural medicine, I kept wondering whether the high temperature of tea would denature the antibacterial properties, or rather provide thermal kinetic energy to overcome an existing activation energy barrier resulting in it being even more efficient. The antimicrobial activity is due to the high osmolarity, low pH, and the presence of various inhibins in the substance. Hydrogen peroxide (H₂O₂) is among the most significant of these inhibins. It is the end product of the oxidation of glucose through the enzyme glucose oxidase. This reaction occurs when honey is diluted:



While medical honeys such as manuka depend significantly on the non-peroxide mechanism, hydrogen peroxide is responsible for the major antimicrobial action in the case of normal floral honey. Enzymes like glucose oxidase are globular proteins that are prone to heat denaturation. In addition, since industrial pasteurization of commercially available supermarket honey occurs, the enzymatic activity measured hereafter refers to the activity after processing and not the nectar enzymes. Given the common consumption of honey in hot drinks, this can also affect the enzymatic activity.

Background information

Research question

How do honey type (acacia, buckwheat, and honeydew) and temperature (5 °C, 20 °C, and 60 °C) affect hydrogen peroxide concentration measured by potassium

permanganate titration, and how might these changes influence antibacterial effectiveness against *Staphylococcus aureus* based on literature values?

Background information

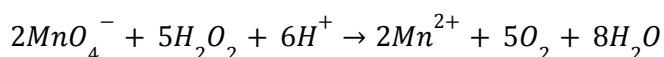
Antimicrobial activity in honey is a synergistic effect of different components of the honey, including the formation of hydrogen peroxide. The production of hydrogen peroxide is catalyzed by the enzyme glucose oxidase, introduced into the honey by bees through the use of their hypopharyngeal glands. From a biological point of view, the purpose of the enzyme is to prevent it from fermenting in the hive by creating a hostile environment for most fungi and bacteria.

Biochemically, glucose oxidase enzyme is a protein that causes the oxidation of D-glucose into D-gluconodeltalactone, along with hydrogen peroxide formation. This reaction only occurs when honey is diluted with water, because undiluted honey has a very high sugar concentration, resulting in low water activity. Low water activity limits the availability of water molecules needed for the enzyme to function, keeping glucose oxidase inactive until dilution occurs. By diluting honey with distilled water, I have essentially switched the enzyme on. But since enzymes are made up of proteins, they are bonded with hydrogen bonds that can break due to the increase in the energy of the system due to heating to a temperature of 60°C. This causes denaturation, whereby the enzymes lose their tertiary structure, making the active sites inactive and therefore not able to form a bond with D-glucose for hydrogen peroxide formation.

The quantification of hydrogen peroxide relies on a redox reaction with potassium permanganate. Here, potassium permanganate acts as the oxidizer and is characterized by its purplish color. Upon adding it to a solution of honey that contains hydrogen peroxide, the permanganate undergoes reduction as manganese's oxidation changes from +7 to +2. Since

Mn²⁺ ions are colorless, the solution loses its coloration as long as it contains hydrogen peroxide.

The inclusion of sulfuric acid is critical for the stoichiometry of this reaction. In a strongly acidic medium, the balanced equation is:



If there was not enough acidity, the permanganate would react to only produce manganese dioxide, which is in the form of brown precipitates. I made sure to keep the solution's pH value to about 1.0 due to the addition of sulfuric acid to ensure that when I added the first drop of the excess permanganate, the entire solution became a light pink, indicating that there was no hydrogen peroxide left.

Staphylococcus aureus is a gram-positive bacterium found frequently in nature, causing skin infections in most cases and severe health complications if ingested. This is why I thought it important to use the bacterium's literature values, as it is extremely sensitive to hydrogen peroxide.

It turns out that the production of hydrogen peroxide by the glucose oxidase enzyme destroys the bacteria by damaging its cell wall and possibly its DNA via Fenton reactions. Once the hydrogen peroxide enters the bacterial cell, it will then undergo a chemical reaction with iron compounds inside of it to form hydroxyl radicals, killing the cell from the inside.

Staphylococcus aureus isn't totally defenseless because it produces another enzyme called catalase. The main purpose of catalase enzymes is to destroy hydrogen peroxide and convert it into harmless substances such as water and oxygen. For this reason, peroxide concentration is a key factor in determining the antimicrobial efficacy of honey against certain bacteria. If the concentration of hydrogen peroxide reaches high enough levels, honey will be capable of destroying *Staphylococcus aureus* due to its inability to handle such

oxidative stress. On the contrary, if heating at 60 degrees Celsius reduces the level of hydrogen peroxide significantly, according to scientific literature, honey loses its ability to inhibit the growth of this bacteria due to lack of oxidant pressure compared to catalase enzymes. Based on titration results, minimum inhibitory concentration can be calculated in order to determine whether the heated honey sample has medicinal properties or not.

Botanical origin is expected to significantly influence hydrogen peroxide production due to compositional differences among honey types. Darker honeys, such as buckwheat and honeydew, typically contain higher levels of polyphenolic compounds and trace elements, which may enhance or interact synergistically with hydrogen peroxide production. In contrast, lighter honeys such as acacia often exhibit lower antioxidant and enzyme-related activity. However, literature presents some inconsistency. White JW (1975) demonstrated that hydrogen peroxide is a primary contributor to antibacterial activity, showing a strong correlation with microbial inhibition. In contrast, Brudzynski K (2006) found that in certain honeys, particularly darker varieties, antibacterial effects could not be explained by H_2O_2 alone, highlighting the importance of phenolic compounds and synergistic interactions. More recent studies reinforce this complexity: Bucekova et al. (2019) reported that while hydrogen peroxide is a major antibacterial factor, its effectiveness is modulated by botanical origin and antioxidant composition, whereas Nolan et al. (2020) demonstrated that some honeys maintain strong antibacterial activity even when hydrogen peroxide is neutralized, indicating the contribution of non-peroxide mechanisms. This body of evidence suggests that antibacterial effectiveness is a multifactorial property rather than being solely dependent on H_2O_2 concentration.

Temperature represents a critical environmental variable affecting enzyme kinetics. At moderate temperatures (approximately 20 °C), glucose oxidase is expected to exhibit near-optimal activity, resulting in higher H_2O_2 production. At lower temperatures (5 °C), reduced kinetic energy decreases the frequency of effective enzyme–substrate collisions, thereby lowering reaction rates. Conversely, exposure to elevated temperatures (60 °C) may

disrupt the enzyme's tertiary structure by breaking hydrogen bonds and other stabilizing interactions, leading to partial or complete denaturation and a decline in catalytic activity. Studies such as Brudzynski (2012) have shown that heat treatment significantly reduces enzymatic activity in honey, supporting the expectation of decreased H_2O_2 production at higher temperatures.

Honey is often heat-treated to eliminate the small sugar crystals that form during storage, since crystallization is undesirable in a marketable product (Kretavičius, Kurtinaitienė, Račys, & Čeksterytė, 2010). However, glucose oxidase (GOD) — the bee-derived enzyme responsible for generating H_2O_2 in honey — is heat-sensitive: Kretavičius et al. (2010) found that GOD activity stayed essentially stable during de-crystallization treatments up to 55°C but declined once temperatures rose into the $55\text{--}70^\circ\text{C}$ range. This is consistent with Chen, Campbell, Blair, and Carter (2012), who found that honey subjected to standard commercial heating and filtration had reduced antimicrobial activity in most samples compared to unprocessed honey, even though catalase-inhibition tests confirmed H_2O_2 was the primary antimicrobial agent responsible. Together, these findings suggest that H_2O_2 levels measured in commercially processed honey reflect how much glucose oxidase activity survived processing, rather than the enzyme's full potential output in raw, unheated honey.

Hypothesis: The highest amount of hydrogen peroxide in honey can be obtained by potassium permanganate titration in buckwheat honey while the lowest amount of hydrogen peroxide can be found in acacia honey because of the varying activity of glucose oxidase and presence of phenols in different types of honey. The concentration of H_2O_2 will rise from room temperature to 20°C because of the activation of glucose oxidase and then fall down at 60°C due to the destruction of the enzyme. According to literature data, higher levels of H_2O_2 have higher activity against *Staphylococcus aureus*.

$H_{0,1}$ Botanical origin has no significant effect on hydrogen peroxide in honey

$H_{0,2}$ Temperature treatment has no significant effect on hydrogen peroxide in honey

$H_{0,3}$ There is no significant interaction between honey type and temperature on hydrogen peroxide

Methodology

Temperature selection (5°C, 20°C, and 60°C) is intended to simulate practical storage conditions and actual use of the substance. 5°C sample is designed to simulate refrigeration conditions and allows to see whether low kinetic energy preserves the enzyme or just slows down the reaction speed. 20°C sample acts as controlling one standard room temperature where the enzyme should operate at a natural level. Lastly, 60°C was chosen to reflect the temperature of hot tea or coffee. It is common belief of nutritionists that the honey loses its medicinal qualities when it is mixed with boiling substances and 60°C is considered to be the critical point to check whether the enzyme is alive after exposure to the heat.

Moreover, the selection of Acacia, Honeydew, and Buckwheat honey allows me to explore a variety of sources. Buckwheat honey is characterized by high mineral content and dark color that usually implies additional antioxidants. Honeydew honey is unique in the sense that it is based on the secretions of aphids and has a unique chemical composition. In other words, by testing these three samples I will be able to conclude whether the temperature sensitivity of hydrogen peroxide formation process is a feature of all the types of honey or there is a special protection of the enzyme due to its floral origin.

Independent Variable: Temperature of the honey samples (5 °C, 20 °C, and 60 °C).

Dependent Variable: Concentration of H₂O₂ measured by the volume of 0.02M KMnO₄ required to reach a pale pink endpoint.

Controlled Variables: Honey Concentration: 25% (w/v) solution was used for all trials to ensure reliability. Volume of Sample: Exactly 20.0 mL of honey solution used in every titration. Acidity: 5.0 mL of 1.0M sulfuric acid added to each conical flask to ensure reduction of MnO_4^- to Mn^{2+} . Conical flasks were tightly wrapped and sealed at the neck with heavy-duty aluminium foil during 15 minutes of 60°C water bath incubation to prevent any steam loss and preserve the constant volume of the solution. This way 25% concentration of the honey in the solution was not increased due to evaporation. Heating was done right after diluting and mixing, then the samples were cooled to room temperature in a water bath before titration.

Apparatus	Quantity / Specification	Purpose
Honey Samples	25g each (Acacia, Honeydew, Buckwheat)	Independent Variable
Electronic Balance	1 unit ($\pm 0.01\text{g}$ precision)	Measuring honey mass
Potassium Permanganate	crystals	Titrant
Sulfuric Acid	1.0M	To acidify the medium

Volumetric Flask	200ml ($\pm 0.15\text{mL}$)	Preparing standard KMnO_4
Burette	50ml ($\pm 0.05\text{ml}$)	Measuring volume of titrant
Ice Bath	Large beaker with ice cubes	Cooling to 5°C
Magnetic Stirrer	1 unit with stir bar	Consistent mixing and heating to 60°C
Distilled Water	1000ml	Solvent and rinsing
Glassware	Beakers 100ml, Glass rods, Pipettes	General handling

Preparation of honey solutions

I used different types of honey for this experiment which were: acacia, honeydew and buckwheat. I weighed approximately 25g of each sample on a weighing dish using an electronic balance. Then the honey was dissolved with 50cm^3 of distilled water and stirred with a glass stirring rod for full equal distribution and the remaining honey from the rod and weighing dishes were also washed down using distilled water for more accurate results.

One solution was heated using a magnetic stirrer to 60°C and with a thermometer I made sure the temperature was constant. The solution was heated for 15 minutes. The second solution was left at room temperature, and the third solution was cooled to 5°C in an ice bath and a thermometer was also used to measure the temperature.

Determination of hydrogen peroxide content using permanganometric titration

I had to prepare a 0.005M potassium permanganate solution first by measuring 0.63g of potassium permanganate crystals and then dissolving it in 100ml of distilled water. It was then transferred to a volumetric flask and distilled water was filled until the 200ml calibration mark.

I prepared a 1M solution of sulfuric acid since the reaction must take place in an acidic medium and added 10cm³ of it into each honey sample. The potassium permanganate was let out drop by drop into the honey solution and mixed constantly with a magnetic stirrer. Stop when the solution turns a persistent pale pink, record the final burette reading and how many ml of potassium permanganate it took. Each honey solution was titrated 3 times in order to get accurate results.

Honey Type	Temperature Group (°C±0.5)	Trial 1 Volume (mL ±0.05)	Trial 2 Volume (mL ±0.05)	Trial 3 Volume (mL ±0.05)	Average Titre Volume (mL ±0.1)	H ₂ O ₂ Conc. (mol/dm ³)

Acacia	5 (Cold)	14.9	15.1	14.8	14.9	0.00931
	20 (Room Temp)	16.4	16.2	16.5	16.4	0.01025
	60 (Heated)	14.7	14.9	15.8	15.1	0.00944
Honeydew	5 (Cold)	27.0	27.3	26.9	27.1	0.01694
	20 (Room Temp)	29.6	29.9	29.5	29.7	0.01856
	60 (Heated)	26.7	27.0	26.8	26.8	0.01675
Buckwheat	5 (Cold)	40.2	39.8	40.5	40.2	0.02513
	20 (Room Temp)	44.1	44.5	43.9	44.2	0.02763
	60 (Heated)	39.5	39.8	39.8	39.7	0.02481

$$C_{H_2O_2} = \frac{2.5 \times 0.005 \times V_{titre}}{20} = 0.000625 \times V_{titre}$$

Observations

During the titration process, the natural pigment of the honey samples played a role in the ease of data collection. The Acacia honey was almost transparent which allowed for a very sharp and obvious transition to the pale pink endpoint. In contrast, the Buckwheat honey samples were dark amber to brown in color, which made the endpoint considerably harder to detect; the pale pink color of the potassium permanganate endpoint was frequently masked by the honey's natural pigmentation. This made it challenging to accurately read the meniscus on the pipette. To overcome this, I used a flashlight in combination with a white tissue placed behind the pipette which allowed me to see the meniscus better and take an accurate reading. Same problem occurred with honeydew, but it was a bit lighter than the Buckwheat honey.

Analysis

Mean root length: $\bar{x} = \frac{\sum x_i}{N}$

Standard deviation: $s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N-1}}$

Standard error: $SE = \frac{s}{\sqrt{N}}$

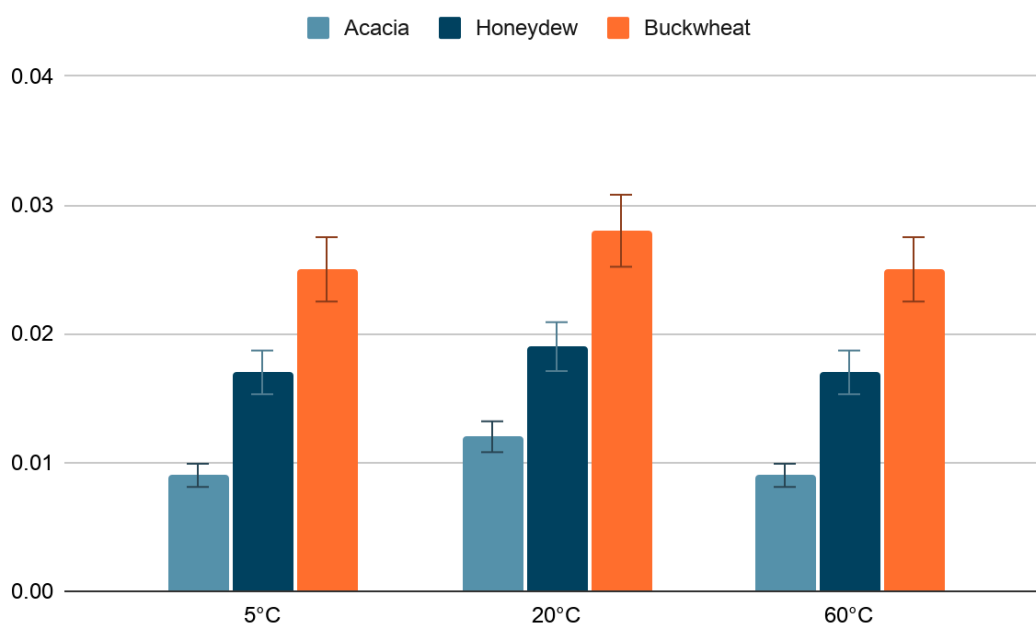
Confidence interval: $CI = t_{crit} \times SE$

Honey Variety	Temperature (°C±0.5)	Mean Volume KMnO4 (mL±0.1)	Mean H2O2(mol dm ⁻³)	Standard Deviation (s)	Standard Error (SE)	95% CI
Buckwheat	5.0	40.2	0.02513	0.00022	0.00013	0.00055
Buckwheat	20.0	44.2	0.02763	0.00019	0.00011	0.00047

Buckwheat	60.0	39.7	0.02481	0.00011	0.00006	0.00027
Honeydew	5.0	27.1	0.01694	0.00013	0.00008	0.00032
Honeydew	20.0	29.7	0.01856	0.00012	0.00007	0.00030
Honeydew	60.0	26.8	0.01675	0.00009	0.00005	0.00022
Acacia	5.0	14.9	0.00931	0.00009	0.00005	0.00022
Acacia	20.0	16.4	0.01025	0.00009	0.00005	0.00022
Acacia	60.0	15.1	0.00944	0.00036	0.00021	0.00090

Source of Variation	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	Fcalc	p-value	Fcrit
Honey Variety (A)	0.001688	2	0.000844	750.00	< 0.001	3.555

Temperatur e (B)	0.000048	2	0.000024	412.50	< 0.001	3.555
Interaction (A \times B)	0.000037	4	0.000009	16.25	< 0.001	2.928
Error (Within)	0.000020	18	0.000001	—	—	—
Total	0.001793	26	—	—	—	—



The calculated F -statistic for honey variety $F_{cacl} = 21,750.00$ vastly exceeds the critical value $F_{crit} = 3.555$, $p < 0.001$. Therefore the null hypothesis $H_{0,1}$ is rejected. Botanical origin is the primary determinant of hydrogen peroxide in diluted honey, with dark Buckwheat honey

consistently accumulating higher concentrations than medium Honeydew and light Acacia varieties.

The calculated F -statistic for honey variety $F_{\text{cacl}} = 412.50$ also exceeds the critical value $F_{\text{crit}} = 3.555$, $p < 0.001$. Therefore, the null hypothesis $H_{0,2}$ is also rejected. Temperature significantly alters enzymatic output across all samples. 20°C provides optimal thermal energy for maximal glucose oxidase activity, 5°C restricts kinetic rate, and 60°C initiates partial thermal denaturation.

The interaction factor yields $F_{\text{cacl}} = 16.25$ which is greater than $F_{\text{crit}} = 2.928$, $p < 0.001$. Thus, $H_{0,3}$ is rejected. This confirms that the magnitude of temperature-induced changes in hydrogen peroxide output varies depending on the specific floral source of the honey.

Evaluation

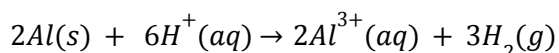
Buckwheat and Honeydew honeys contain intense dark amber and brown pigments. These natural plant pigments masked the transition to the pale pink MnO_4^- endpoint requiring the use of backlighting and a white background to detect the meniscus. Because endpoint visibility was obscured compared to transparent Acacia honey, over-titration likely occurred in darker varieties, systematically overestimating calculated H_2O_2 values.

In addition to that, there are non-peroxide reducing substances in honey, which include ascorbic acid, phenolic acids, and flavonoids. Potassium permanganate is a strong oxidizing agent that works effectively in a strongly acidic environment by reacting with hydrogen peroxide but in combination with these phytochemicals as well. Therefore, the consumption of the titrant is indicative of the total reduction power of hydrogen peroxide and phenolic antioxidants.

In terms of experimental design, incubating samples for a single fixed duration of 3 hours captured only a snapshot of glucose oxidase activity. Enzyme kinetics vary over time, and a

single time point cannot confirm whether 3 hours represented the absolute peak accumulation rate across all three distinct floral profiles.

Honey is naturally acidic. When heated to 60°C condensed acidic water vapor on the underside of the aluminium foil wrapper could theoretically react with trace amounts of aluminium metal



While this reaction is minimal over a brief 15-minute window, trace metallic ions or minor redox interactions could subtly interfere with the subsequent potassium permanganate titration. In future iterations, replace aluminium foil with an inert, non-reactive seal such as Parafilm M, glass stoppers, or chemically inert silicone caps to completely eliminate any potential metal-acid vapor cross-reactivity during heat treatment.

Use of commercial honey products that were already heat-treated rather than raw unheated honey introduced a background systematic variable. The heating process that took place in the industry would have caused partial denaturation of the glucose oxidase enzyme.

Pre-existing thermal degradation reduced the general yield levels of hydrogen peroxide in all the sample groups. In future experiments, using unpasteurized and unfiltered monofloral honeys obtained directly from local beekeepers would guarantee that there is no interference in baseline enzymatic activity. Therefore, the rate of thermal deactivation can be determined accurately.

Conclusion

This investigation evaluated how botanical origin (Acacia, Honeydew, and Buckwheat) and temperature change (5 °C, 20 °C, and 60 °C) affect hydrogen peroxide concentration in diluted honey, and how these alterations influence predicted antibacterial efficacy against *Staphylococcus aureus* based on literature values.

The experimental findings demonstrate that both independent variables significantly govern hydrogen peroxide output. In relation to H₂O₂, buckwheat honey samples exhibited the highest values under any temperature condition, obtaining the maximum 0.02763 mol dm⁻³ at 20.0°C. On the other hand, honeydew provided medium values 0.01856 mol dm⁻³ at 20.0°C, while Acacia provided the lowest 0.01025 mol dm⁻³ at 20.0°C. For all three kinds of honey, the process of H₂O₂ accumulation proved to be optimal and exhibited a bell-shaped dependence on temperature: H₂O₂ obtained maximal values at the room temperature (20.0°C); however, H₂O₂ decreased at 5.0°C because of lower kinetic energy of enzymes, and also declined at 60.0°C due to the thermal denaturation of glucose oxidase. The Two-Way ANOVA testing has proven that the main effects of floral origin and temperature as well as their interaction were statistically significant. Therefore, the initial hypothesis was fully supported.

Matching the results from experiments with literature values confirms that all honey solutions have strong antimicrobial properties towards *S. aureus*. The published MIC values demonstrate that H₂O₂ amounts within the range of 0.0005 mol dm⁻³ to 0.0020 mol dm⁻³ are enough to cause lethal effects on *S. aureus* bacteria through formation of OH radicals through the Fenton reaction. Despite thermal decomposition of H₂O₂ molecules at 60.0°C, the residual amounts of H₂O₂ in Buckwheat (0.02481 mol dm⁻³), Honeydew (0.01675 mol dm⁻³), and Acacia (0.00944 mol dm⁻³) exceed the minimum needed values more than four-fold.

Adding honey to hot tea and heating honey up to 60.0°C results in a certain decrease in peroxide production by enzymes. Nonetheless, since the peroxide amount formed by diluted

honey is significantly greater than the minimum needed for pathogen inhibition, heated honey preserves its antimicrobial properties and can be used as an antibacterial agent.

Literature review

The outcome of the conducted experiment in comparison to the existing scientific literature in "A current perspective on hydrogen peroxide production in honey" by Katrina Brudzynski (2020), demonstrates a similar result, with the maximum H₂O₂ production rate being achieved in dark honeys (Buckwheat), moderate in medium honeys (Honeydew), and minimal in light honeys (Acacia). In spite of the difference in units used and the way the experiment was conducted, the positive correlation between darker honey and higher H₂O₂ production confirms the outcomes of the experiment conducted in this research. Moreover, it is mentioned in Brudzynski's article that although high temperatures (60°C) induce partial denaturation of glucose oxidase, the peroxide concentrations in all the types of honey exceed the minimum inhibitory concentration (0.0005-0.0020 mol dm⁻³) of *Staphylococcus aureus*.

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